

End-to-End Project Validation Checklist

End-to-End Validation Checklist

Status: Validation package created. Hardware validation **NOT TESTED**.

Target: Full VelaDial system: Raspberry Pi + Home Assistant + LocalTuya/bedroom lights + door-side ELECROW ESP32-S3 + bedside ESP32-C6/APDS-9960.

Rule: Do not mark any row PASS until Hardik has physically performed the step and captured evidence.

This is the technician-style field runbook for validating VelaDial from zero setup to final PASS/FAIL sign-off. It intentionally separates **setup**, **subsystem validation**, **full end-to-end validation**, **failure triage**, and **evidence capture**.

How to use this checklist

1. Complete stages in order. Do not skip ahead.
2. Mark each row as one of: **PENDING**, **PASS**, **FAIL**, **N/A**, or **RETEST REQUIRED**.
3. For every **FAIL**, record the failure in `end_to_end_test_results_template.md` before changing firmware or configuration.
4. Do not edit production firmware during validation unless a failure is confirmed and a new fix PR is opened.
5. Hardware PASS requires every required stage to pass on physical hardware. Compile success alone is not hardware PASS.

Required result labels

LABEL	MEANING
PENDING	Not tested yet. Default state.
PASS	Tested on real hardware and met expected behavior.

LABEL	MEANING
FAIL	Tested and did not meet expected behavior. Must include evidence and reproduction steps.
N/A	Not applicable to this hardware setup. Explain why.
RETEST REQUIRED	Previously failed or changed; must be tested again.

Stop/go gates

GATE	MUST PASS BEFORE CONTINUING	IF FAILED
Gate A — HA reachable	Home Assistant dashboard loads from phone/ computer	Stop. Fix Raspberry Pi/ network first.
Gate B — lights controllable in HA	<code>light.bedroom_group</code> works from HA UI before ESP devices are tested	Stop. Fix bulbs/LocalTuya/ group mapping first.
Gate C — door-side boots	ELECROW boots, display renders, ESPHome API connects	Stop. Fix flash/pin/display/ network before UI testing.
Gate D — long-press safe	Long press opens Theme Selector and does not toggle lights	Stop. Do not proceed until fixed. This is a safety gate.
Gate E — bedside boots	ESP32-C6 boots, APDS-9960 is detected, ESPHome API connects	Stop. Fix wiring/flash/I2C before gesture testing.
Gate F — full E2E path	Door-side and bedside both control <code>light.bedroom_group</code> through HA/local integration	Only then consider conditional hardware PASS.

Stage 0 — Required hardware and materials

ITEM	REQUIRED DETAIL	STATUS	EVIDENCE / NOTES
Raspberry Pi	Model and power supply known; stable 5V supply.	PENDING	
microSD card	HA OS target card available; capacity recorded.	PENDING	

ITEM	REQUIRED DETAIL	STATUS	EVIDENCE / NOTES
Network access	Same LAN for Pi, ESP devices, phone/computer, and bulbs.	PENDING	
Computer or phone	Can access Home Assistant web UI.	PENDING	
ELECROW CrowPanel 1.28" ESP32-S3 rotary display	Door-side hardware present.	PENDING	
Adafruit ESP32-C6 Feather	Bedside controller present.	PENDING	
APDS-9960 gesture sensor	Connected via STEMMA/Qwiic or wired I2C.	PENDING	
USB cables	Data-capable cables for both boards.	PENDING	
Bedroom smart bulbs / Tuya bulbs	Installed, powered, reset/ onboarded as needed.	PENDING	
Required credentials	Wi-Fi, HA account, Tuya/local keys available privately.	PENDING	No secrets committed to repo.
Repo checkout	Latest <code>main</code> pulled after PR #57 merge.	PENDING	Commit SHA:

Stage 1 — Raspberry Pi Home Assistant setup

1.1 Flash and boot Home Assistant OS

STEP	ACTION	EXPECTED RESULT	STATUS	EVIDENCE / NOTES
1.1.1	Use Raspberry Pi Imager to flash Home Assistant OS to microSD.	Flash completes without error.	PENDING	
1.1.2	Insert microSD, connect Ethernet/Wi-Fi as applicable, power Pi.	Pi powers on and boots.	PENDING	
1.1.3	Wait for first boot and onboarding.	HA becomes reachable after initial setup time.	PENDING	

STEP	ACTION	EXPECTED RESULT	STATUS	EVIDENCE / NOTES
1.1.4	Open <code>http://homeassistant.local:8123</code> or Pi IP address.	HA onboarding page loads.	PENDING	
1.1.5	Create Home Assistant owner account.	Dashboard loads after sign-in.	PENDING	

1.2 Failure triage — Raspberry Pi / HA access

SYMPTOM	CHECK FIRST	NEXT ACTION	STATUS
No HA URL	Router client list / IP scan	Try IP:8123 directly.	PENDING
Pi not booting	Power supply, SD card, HDMI/LED behavior	Reflash SD card if needed.	PENDING
HA slow/unreachable	Wait for initial boot, check network	Use Ethernet for first setup if possible.	PENDING
Browser cannot connect	Same network/VPN off	Confirm phone/computer is on same LAN.	PENDING

Stage 2 — Home Assistant base configuration

2.1 ESPHome add-on setup

STEP	ACTION	EXPECTED RESULT	STATUS	EVIDENCE / NOTES
2.1.1	Install ESPHome add-on in HA.	Add-on installs successfully.	PENDING	
2.1.2	Start ESPHome add-on.	ESPHome dashboard opens.	PENDING	
2.1.3	Confirm ESPHome can access/import project YAMLs.	Door-side and bedside YAMLs are visible or uploadable.	PENDING	
2.1.4			PENDING	

STEP	ACTION	EXPECTED RESULT	STATUS	EVIDENCE / NOTES
	Confirm no real secrets are in repo files.	Credentials exist only in local <code>secrets.yaml</code> or HA secrets.		

2.2 LocalTuya / bedroom light setup

STEP	ACTION	EXPECTED RESULT	STATUS	EVIDENCE / NOTES
2.2.1	Install/configure LocalTuya or chosen local bulb integration.	Integration loads without errors.	PENDING	
2.2.2	Add each bedroom bulb.	Bulbs appear in HA as controllable light entities.	PENDING	
2.2.3	Validate each bulb individually.	ON/OFF, brightness, and color temperature work from HA UI.	PENDING	
2.2.4	Create <code>light.bedroom_group</code> .	Group entity appears in HA.	PENDING	
2.2.5	Test <code>light.bedroom_group</code> ON/OFF.	All intended bedroom bulbs respond.	PENDING	
2.2.6	Test <code>light.bedroom_group</code> brightness.	All bulbs dim/brighten consistently.	PENDING	
2.2.7	Test <code>light.bedroom_group</code> color temperature.	Warm/cool changes work as expected.	PENDING	

2.3 LocalTuya / group failure triage

SYMPTOM	LIKELY AREA	CHECK	STATUS
Bulb unavailable	Network/local key/device ID	Verify bulb IP, local key, Tuya device mapping.	PENDING
ON/OFF works but brightness fails	DP mapping	Verify brightness DP and scale.	PENDING
Color temperature reversed/wrong	DP mapping/range	Verify CT min/max and DP mapping.	PENDING

SYMPTOM	LIKELY AREA	CHECK	STATUS
Group does not respond	Group entity config	Confirm group members and entity ID exactly <code>light.bedroom_group</code> .	PENDING

Gate B: Do not test ESP devices until `light.bedroom_group` works from HA UI.

Stage 3 — Door-side ELECROW flashing

3.1 Compile and flash

STEP	COMMAND / ACTION	EXPECTED RESULT	STATUS	EVIDENCE / NOTES
3.1.1	Pull latest repo main after PR #57 merge.	Local repo is current.	PENDING	Commit SHA:
3.1.2	Verify local secrets are present but not committed.	<code>esphome/secrets.yaml</code> exists locally only.	PENDING	
3.1.3	Compile door-side firmware.	<code>esphome compile esphome/door_side_rotary.yaml</code> passes.	PENDING	
3.1.4	Connect ELECROW board via USB-C data cable.	Board appears as serial device.	PENDING	Port:
3.1.5	Flash via USB first.	<code>esphome run esphome/door_side_rotary.yaml</code> uploads successfully.	PENDING	
3.1.6	Keep serial monitor open after reboot.	No bootloop, panic, or repeated reset.	PENDING	

3.2 Door-side first boot checkpoint

TEST ITEM	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
	Turns on after boot.	PENDING	

TEST ITEM	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
Display backlight			
UI render	Power page or expected default page appears.	PENDING	
Wi-Fi	Device joins configured Wi-Fi.	PENDING	
HA API	Device appears online in Home Assistant / ESPHome.	PENDING	
Active theme entity	<code>text_sensor.veladial_doorside_active_theme</code> appears if implemented.	PENDING	
Serial logs	No critical errors for display, touch, sensors, or HA API.	PENDING	

Stage 4 — Door-side physical validation

4.1 Input and wake-only-first validation

TEST ITEM	PROCEDURE	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
Screen sleep	Leave device idle until timeout.	Screen dims/sleeps as designed.	PENDING	
Touch wake-only	While asleep, tap/touch once.	Wakes only; no toggle, preset, or page action.	PENDING	
Knob rotation wake-only	While asleep, rotate one detent.	Wakes only; brightness/preset does not change.	PENDING	
Knob press wake-only	While asleep, press knob once.	Wakes only; light does not toggle.	PENDING	
Short press	While awake, short press on Power page.	Toggles <code>light.bedroom_group</code> .	PENDING	
Long press	Hold knob >1.5s.	Opens Theme Selector.	PENDING	
Long press isolation	Watch bulbs while long-pressing.	Long press does not toggle or apply action.	PENDING	Safety gate.

TEST ITEM	PROCEDURE	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
Swipe left/right	Swipe pages while awake.	Navigates Power / Brightness / Presets.	PENDING	
Page indicator	Navigate pages.	Indicator matches active page.	PENDING	

4.2 Control validation

TEST ITEM	PROCEDURE	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
Power page	Short press or tap Power.	<code>light.bedroom_group</code> toggles ON/OFF.	PENDING	
Brightness page	Rotate knob CW/CCW.	Brightness changes in visible steps.	PENDING	
Brightness HA sync	Watch HA light entity during changes.	HA reflects brightness value.	PENDING	
Preset selection	Rotate on Presets page.	Highlight moves through 4 presets.	PENDING	
Warm White	Apply preset.	Light changes to Warm White target.	PENDING	
Soft Amber	Apply preset.	Light changes to Soft Amber target.	PENDING	
Neutral White	Apply preset.	Light changes to Neutral White target.	PENDING	
Low Nightlight	Apply preset.	Light changes to Low Nightlight target.	PENDING	

4.3 Theme Selector validation

TEST ITEM	PROCEDURE	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
Open selector	Long press knob.	Theme Selector opens.	PENDING	

TEST ITEM	PROCEDURE	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
Browse forward	Rotate CW.	Theme preview index increments.	PENDING	
Browse backward	Rotate CCW.	Theme preview index decrements.	PENDING	
Wrap behavior	Rotate past first/last.	Selector wraps or stops as documented.	PENDING	
Apply theme	Short press selected theme.	Theme applies and returns to main UI.	PENDING	
Persist theme	Power-cycle/reboot after applying.	Same theme remains active.	PENDING	
HA theme sensor	Check HA entity.	Active theme text updates after apply.	PENDING	

4.4 Theme-by-theme visual validation

#	THEME	MINIMUM EXPECTED VISIBLE DIFFERENCE	STATUS	VISUAL ISSUE / NOTES
1	Minimal Thermostat	Minimal thermostat-style palette/ labels/motif.	PENDING	
2	SmartKnob-Inspired Arc	Arc/knob-inspired color and control emphasis.	PENDING	
3	Large Center Power Button	Large central power emphasis.	PENDING	
4	Single-Page Simple Mode	Simplified/low-density presentation.	PENDING	
5	Preset Ring UI	Preset/ring-inspired visual treatment.	PENDING	
6	Night Mode Ultra-Minimal	Very dim night-safe palette.	PENDING	
7	Text-First Utility	Typography-first presentation.	PENDING	
8			PENDING	

#	THEME	MINIMUM EXPECTED VISIBLE DIFFERENCE	STATUS	VISUAL ISSUE / NOTES
	Apple Watch Complications	Small status/complication-inspired treatment.		
9	LED-Ring Status-First	LED ring is visually important.	PENDING	
10	Three-Screen Tab Carousel	Clear tab/page-carousel identity.	PENDING	
11	Brightness-First UI	Brightness is the default hero.	PENDING	
12	Door Switch Replacement	Obvious wall-switch mental model.	PENDING	
13	Lunar Phase Visualization	Moon/lunar visual identity.	PENDING	
14	Sundial Shadow UI	Light/shadow/sundial identity.	PENDING	
15	Tree Ring Growth Pattern	Organic/tree-ring identity.	PENDING	
16	Topographic Contour Map	Contour/elevation identity.	PENDING	
17	Iris Aperture Mechanism	Optical aperture identity.	PENDING	
18	Radar Sweep Animation	Scanning/sweep identity.	PENDING	
19	Vinyl DJ Crossfader	Turntable/audio-instrument identity.	PENDING	
20	Eclipse Corona	Eclipse/corona glow identity.	PENDING	

4.5 Sensor, LED, and unavailable-state validation

TEST ITEM	PROCEDURE	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
LED ring ON	Turn lights ON.	LED ring follows active theme behavior.	PENDING	
LED ring OFF	Turn lights OFF.	LED ring dims/off as designed.	PENDING	

TEST ITEM	PROCEDURE	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
Theme LED color	Apply several themes.	LED color/behavior changes with theme.	PENDING	
LED brightness comfort	Observe in dark room.	Not painfully bright.	PENDING	
TSL2591 adaptive backlight	Change room light level if sensor present.	Display backlight changes appropriately.	PENDING	
SHT45 diagnostic	Check HA diagnostics if sensor present.	Temp/humidity present; not a main UI page.	PENDING	
API unavailable	Temporarily disconnect HA/network safely.	UI shows unavailable state without crashing.	PENDING	

Stage 5 — Bedside ESP32-C6 flashing

STEP	COMMAND / ACTION	EXPECTED RESULT	STATUS	EVIDENCE / NOTES
5.1	Confirm APDS-9960 wiring / STEMMA connection.	Sensor physically connected.	PENDING	
5.2	Compile bedside YAML.	<code>esphome compile esphome/bedside_gesture.yaml</code> passes.	PENDING	
5.3	Connect ESP32-C6 via USB data cable.	Serial device appears.	PENDING	Port:
5.4	Flash via USB.	<code>esphome run esphome/bedside_gesture.yaml</code> uploads successfully.	PENDING	
5.5	Keep serial monitor open after reboot.	No bootloop, panic, or repeated reset.	PENDING	
5.6	Confirm Wi-Fi/API connection.	Device appears online in HA / ESPHome.	PENDING	
5.7		No I2C/sensor initialization errors.	PENDING	

STEP	COMMAND / ACTION	EXPECTED RESULT	STATUS	EVIDENCE / NOTES
	Confirm APDS-9960 detection.			

Stage 6 — Bedside gesture validation

TEST ITEM	PROCEDURE	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
LEFT gesture	Swipe left above APDS-9960.	<code>light.bedroom_group</code> turns OFF.	PENDING	
RIGHT gesture	Swipe right above APDS-9960.	<code>light.bedroom_group</code> turns ON.	PENDING	
Cooldown	Repeat gestures quickly.	Cooldown prevents rapid repeated triggers.	PENDING	
False trigger check	Stand nearby / move normally.	No accidental triggers.	PENDING	
HA reflects state	Watch <code>light.bedroom_group</code> .	HA state updates after gesture.	PENDING	
APDS diagnostics	Check available entities/ logs.	Sensor status is stable.	PENDING	
VL53L4CD v1 behavior	Try hand-hold if hardware attached.	No v1 action unless explicitly implemented later.	PENDING	
Sensor fusion	Review behavior/logs.	No APDS+VL53 fusion behavior in v1.	PENDING	

Stage 7 — Full end-to-end system validation

TEST ITEM	PROCEDURE	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
	Use door Power page.		PENDING	

TEST ITEM	PROCEDURE	EXPECTED BEHAVIOR	STATUS	EVIDENCE / NOTES
Door → HA → lights power		Bedroom group toggles; HA state matches.		
Door → HA → lights brightness	Use door Brightness page.	Bulbs change brightness; HA state matches.	PENDING	
Door → HA → presets	Apply all 4 presets.	Bulbs match expected preset behavior.	PENDING	
Door theme persistence	Apply theme, reboot door device.	Same theme returns.	PENDING	
Bedside → HA → lights OFF	LEFT gesture.	Bedroom group turns OFF.	PENDING	
Bedside → HA → lights ON	RIGHT gesture.	Bedroom group turns ON.	PENDING	
Cross-device state recovery	Change lights from one device, observe other path.	HA remains source of truth; no stuck state.	PENDING	
HA dashboard reflects both	Perform door and bedside actions.	HA UI updates for both.	PENDING	
Local-only behavior	Disconnect internet but keep LAN if safe.	Local HA/LocalTuya path continues if configured locally.	PENDING	
Pi reboot recovery	Reboot Raspberry Pi.	ESP devices reconnect; controls recover.	PENDING	
Door reboot recovery	Reboot ELECROW board.	Device reconnects and theme persists.	PENDING	
Bedside reboot recovery	Reboot ESP32-C6.	Device reconnects; gestures resume.	PENDING	
30-minute soak	Leave system running and test intermittently.	No crashes, disconnect loops, overheating, or UI lockups.	PENDING	

Stage 8 — Evidence collection

EVIDENCE ITEM	REQUIRED FOR PASS?	STATUS	LINK / FILENAME / NOTES
Photo of HA dashboard loaded on Raspberry Pi system	Yes	PENDING	
Screenshot/photo of ESPHome devices online	Yes	PENDING	
Screenshot of <code>light.bedroom_group</code> entity	Yes	PENDING	
Door-side boot photo/video	Yes	PENDING	
Video: long press opens Theme Selector	Yes	PENDING	
Video: long press does not toggle light	Yes	PENDING	
Video: browsing all 20 themes	Yes	PENDING	
Photo/video: selected theme persists after reboot	Yes	PENDING	
Video: door controls light group	Yes	PENDING	
Video: bedside LEFT/RIGHT gesture controls lights	Yes	PENDING	
Serial logs for any failure	Required if failure occurs	PENDING	
Final completed results template	Yes	PENDING	

Stage 9 — Final system sign-off

SUBSYSTEM	REQUIRED RESULT BEFORE SYSTEM PASS	RESULT	EVIDENCE LINK / NOTES
Raspberry Pi setup	PASS	PENDING	
Home Assistant setup	PASS	PENDING	
LocalTuya / <code>light.bedroom_group</code> setup	PASS	PENDING	

SUBSYSTEM	REQUIRED RESULT BEFORE SYSTEM PASS	RESULT	EVIDENCE LINK / NOTES
Door-side flashing	PASS	PENDING	
Door-side hardware validation	PASS or conditional PASS with documented limits	PENDING	
Bedside flashing	PASS	PENDING	
Bedside gesture validation	PASS or conditional PASS with documented limits	PENDING	
Full E2E command path	PASS	PENDING	
Evidence package complete	PASS	PENDING	
Final decision	☐ PASS / ☐ FAIL / ☐ ☐ CONDITIONAL PASS	PENDING	

Conditional PASS rules

A conditional PASS is allowed only when the core system works and limitations are explicitly documented. Examples:

CONDITION	CONDITIONAL PASS ALLOWED?	NOTES
One visual theme needs cosmetic refinement but controls work	Yes	Log visual issue and continue.
TSL2591 not present on board revision but core UI works	Yes, if documented	Mark TSL2591 as N/A for that hardware revision.
LED ring color order is wrong but controls work	Yes, if documented	Requires follow-up firmware fix before final polish.
Long press also toggles light	No	Safety gate failure.
<code>light.bedroom_group</code> cannot be controlled from HA	No	E2E gate failure.
Door-side firmware bootloops	No	Core hardware gate failure.
Bedside gestures cannot control lights	No for full system PASS	Can only pass door-side subsystem.

Final note

This checklist is intentionally strict. VelaDial is not physically validated until the filled results template contains evidence-backed PASS/FAIL results from real hardware testing.